Seasonality Anomalies in the US Equity Market

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Seasonality anomalies

- Bouman and Jacobsen (2002) documented Sell in May and go away, with November to April returning materially more than May to October in most markets they examined.
- Rozeff and Kinney (1976) first documented the January effect for US equities, with prices rising more in January than in any other month.
- Ariel (1987) documented abnormally high returns around the boundary between calendar months, the turn-of-month effect.
- Cooper et al. (2006) studied the January barometer, which holds that the January return predicts the return over the rest of the year.
- McLean and Pontiff (2016) found that return predictability decays substantially after publication.

Research question

Question

What remains of seasonality anomalies in the US equity market, and has trading on them become less profitable?

Finding

1. November to April exceeds May to October by 0.81% a month.
2. The January effect is not stable, and the January barometer is rejected.
3. The Halloween and turn-of-month strategy matches buy and hold with dividends on return, volatility and Sharpe ratio, and slightly improves on its drawdown.

The data are daily S&P 500 closing levels and their log returns from 1981 to 2023, with the 3-month Treasury bill secondary market rate as the risk-free rate.

Seasonal decomposition

A classical additive decomposition writes the return as a trend, a seasonal component and a residual. The seasonal component at each calendar position is the mean of the detrended series at that position.

- Monthly factor

$$\hat{s}_m = \frac{1}{|Y_m|} \sum_{y \in Y_m} \left(R_{y,m} - \bar{R}_y \right), \qquad m = 1, \ldots, 12,$$

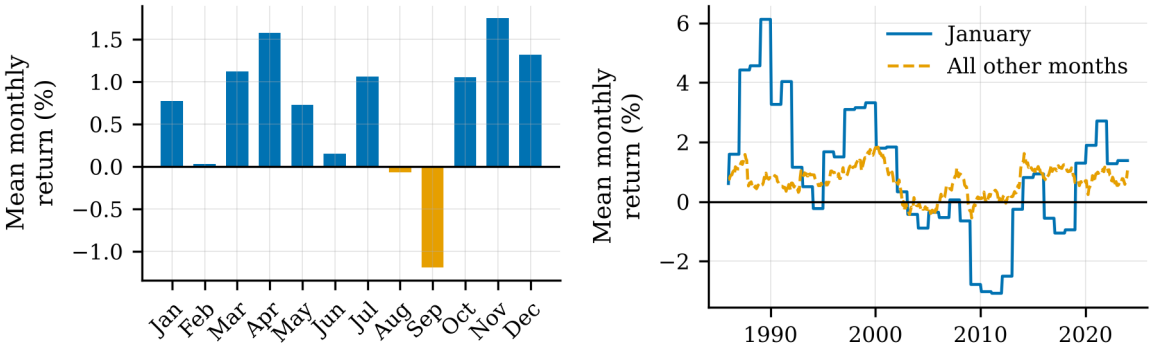
with $R_{y,m}$ the return of month m in year y, $\bar{R}_y$ the mean monthly return in year y and Y_m the years in which month m is observed.

- Daily factor Months contain between 15 and 23 trading days in this sample, so day j counts forward from the first trading day and backward from the last,

$$\hat{s}_j^{+} = \underset{(y,m)}{\text{mean}} \left(r_{y,m,j} - \bar{r}_{y,m} \right), \qquad \hat{s}_j^{-} = \underset{(y,m)}{\text{mean}} \left(r_{y,m,\, n_{y,m}-j+1} - \bar{r}_{y,m} \right),$$

with $r_{y,m,j}$ the day j return of month (y, m), $n_{y,m}$ its trading days and $\bar{r}_{y,m}$ its daily mean.

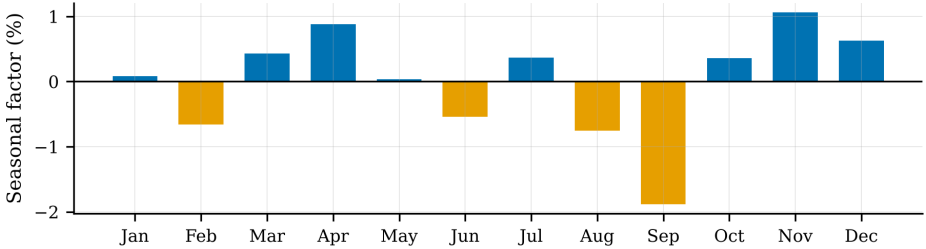
Monthly seasonality



Mean monthly return and the 5-year rolling average of January against all other months.

- January returns on average 0.77% a month, slightly above the sample mean.
- The rolling comparison changes sign in half of all windows, so the January effect is not stable in this sample.

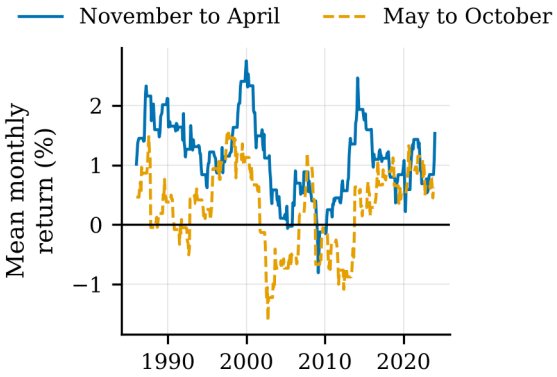
Monthly seasonal factor

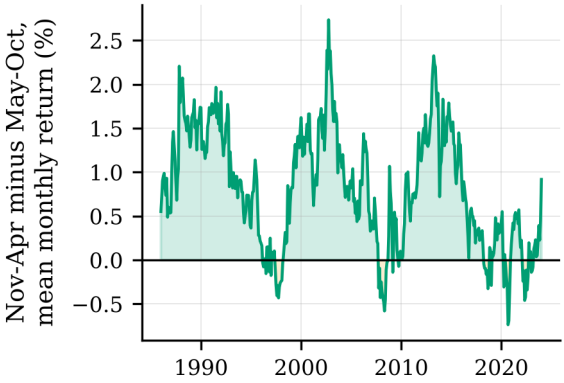


Estimated monthly seasonal factor.

- November at $+1.06\%$ and April at $+0.88\%$ are the two strongest months, both inside the November to April window.
- September at -1.88% is by a wide margin the weakest. The pattern is primarily driven by a weak September and a strong year-end.

Halloween effect

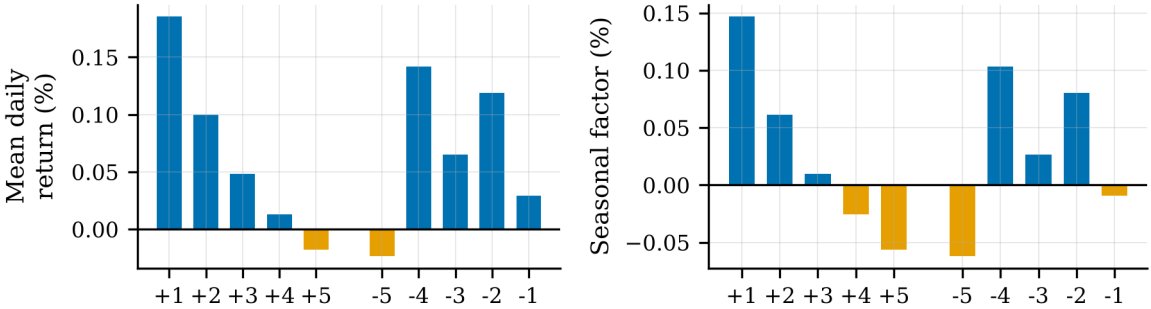




5-year rolling average monthly return in each half-year, and the difference between them.

- The November to April average exceeds the May to October average by 0.81% a month.
- The gap is negative in only 12.5% of 5-year windows.

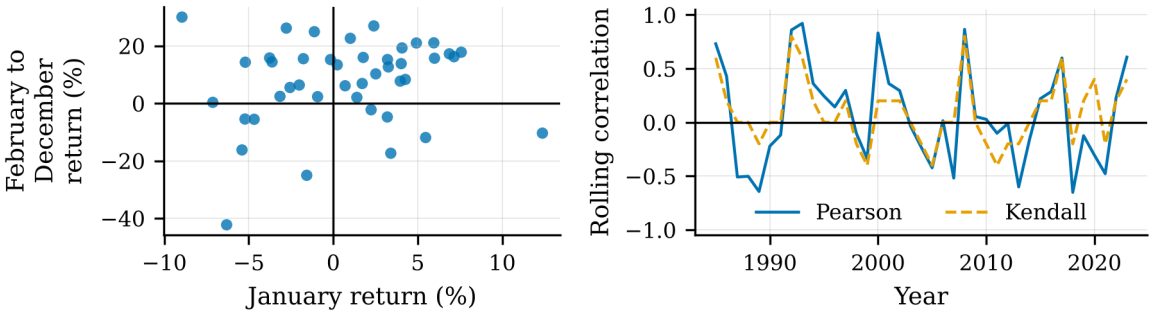
Daily seasonality



Mean daily return and estimated daily seasonal factor over the first and the last 5 trading days.

- The first trading day carries a factor of $+0.147\%$ and the second $+0.061\%$, while the fourth and fifth are negative.
- The final trading day is indistinguishable from zero at -0.010%, but the fourth, third and second to last are all positive.

January barometer



5-year rolling correlation between the January return and the February to December return.

- The Pearson correlation between the January and the February to December return is $+0.156$ across 43 years, not statistically significant.
- Pearson and Kendall correlations change sign repeatedly, so the barometer is rejected.

Trading strategies

- S1 Halloween Buy at the close of the last trading day of October, sell at the close of the last trading day of April, and hold Treasury bills for the rest of the year.
- S2 Halloween and turn-of-month S1 plus the index over the first and the last 5 trading days of each month, levered to 1.5 where the two signals coincide.
- P1 Buy and hold The index held throughout the sample, excluding dividends.
- P2 Buy and hold with dividend P1 credited with the sample average annual dividend yield of 2.48%, accrued evenly across trading days.

The additional half of a levered position is funded at the bill rate, and the backtest charges 5 basis points one-way on every change in position.

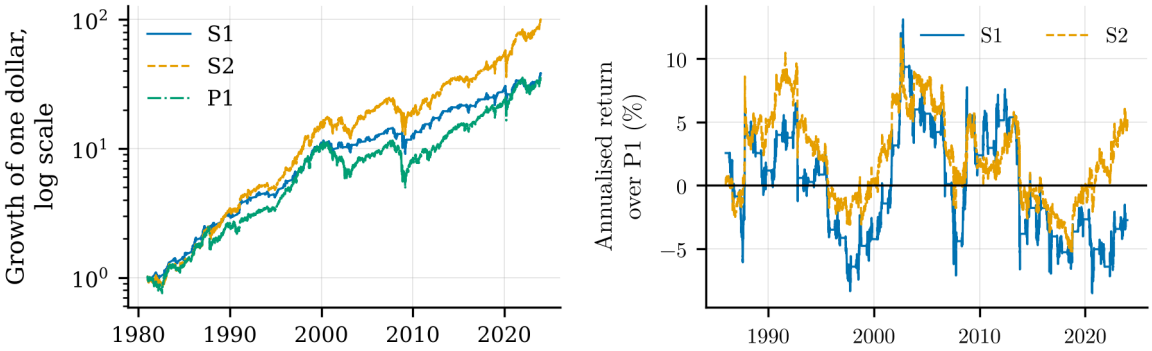
Backtest result

Strategy	Return (%)	Volatility (%)	Sharpe	Sortino	Max DD (%)
S1 Halloween	8.48	12.5	0.360	0.508	−37.2
S2 Halloween and turn-of-month	10.7	17.9	0.377	0.533	−48.2
P1 Buy and hold	8.27	18.1	0.238	0.328	−56.8
P2 Buy and hold with dividend	10.8	18.1	0.376	0.520	−55.2

Performance from 1981 to 2023. Returns and volatility are annualized.

- Both strategies outperform P1 on every measure.
- Neither exceeds the total return benchmark P2 on return, and their Sharpe ratios sit within 0.02 of it.
- S2 has very close return, volatility and Sharpe ratio to P2, and its maximum drawdown is 7% shallower.

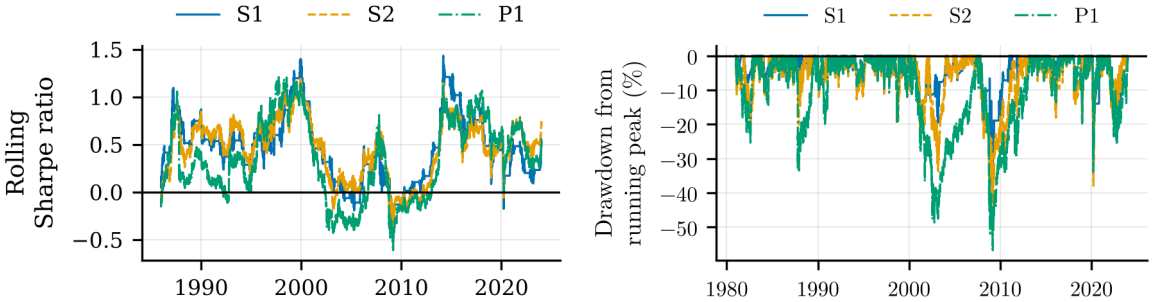
Growth and rolling outperformance



Growth performance under S1, S2 and P1, and the 5-year rolling outperformance over P1.

- The 5-year rolling outperformance of S2 averages $+2.38\% \to +2.25\% \to +2.03\%$ before 2000, from 2000 to 2019 and from 2020.
- It is not statistically significant.

Rolling Sharpe ratio and drawdown



5-year rolling Sharpe ratio, and drawdown from running peak.

- The rolling Sharpe ratio of S2 leads P1 by 0.153 → 0.098 → 0.048 before 2000, from 2000 to 2019 and from 2020, a monotone narrowing.
- S1 is shallower than P1 on 86% of days and S2 on 63%, with the gap widest in the dot-com and global financial crises.

Discussion

- We analyze monthly and daily seasonality in the US equity market and backtest the Halloween and turn-of-month strategies from 1981 to 2023.
- S2 matches P2 on return, volatility and Sharpe ratio and improves on its drawdown.
- Its rolling outperformance over P1 in return and Sharpe ratio decreases over time and is not statistically significant.
- We cannot conclude that there was alpha in trading seasonality anomalies in the past, nor that there is alpha decay. Charging 5 basis points and neglecting slippage, market impact and taxes may understate historical costs, which would inflate the early outperformance and any apparent decay.

References

- Ariel, R. A. (1987). A monthly effect in stock returns. *Journal of Financial Economics*, 18(1), 161–174.
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Code and replication data are available at *GitHub*.